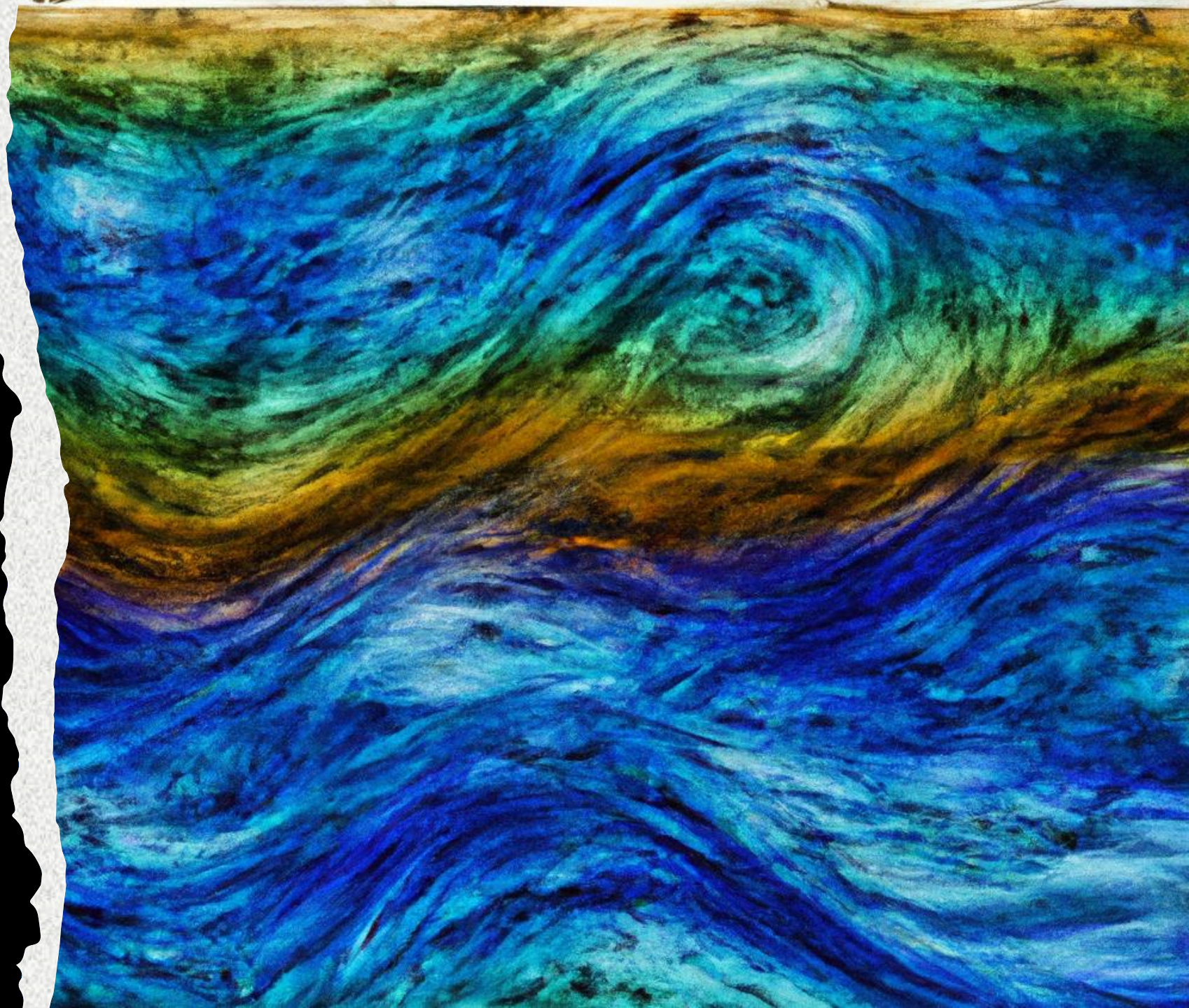
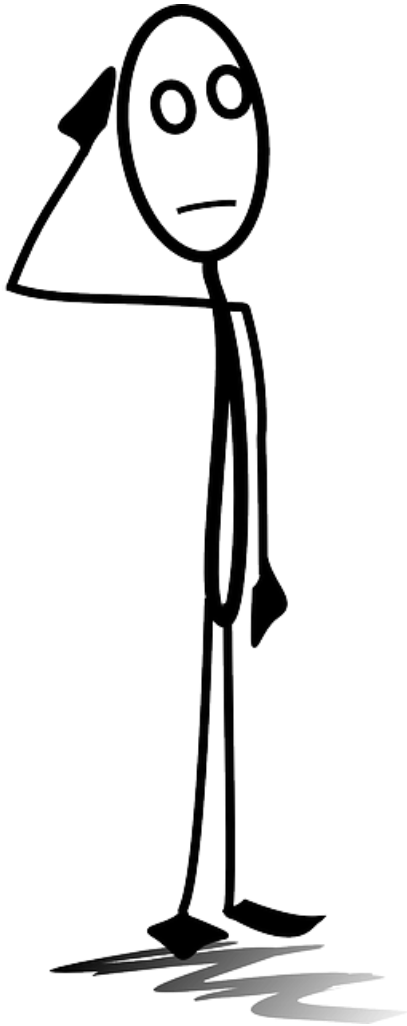
PHYSICS I
WEEK 7

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Office: HSB 116B



WEET 7: OUTLINE

- Review: Phase portraits
- Bounded/Unbounded motion (Thornton-Marion 4.3,4.4)



REVIEW: LINEAR 1D SYSTEMS

1. Find $U(x) \rightarrow$

$$U(x) = -\int_{x_0}^{x} F_x(x')dx'$$

2. Find turning points $x_{\pm}$where

$$U(x) = E$$

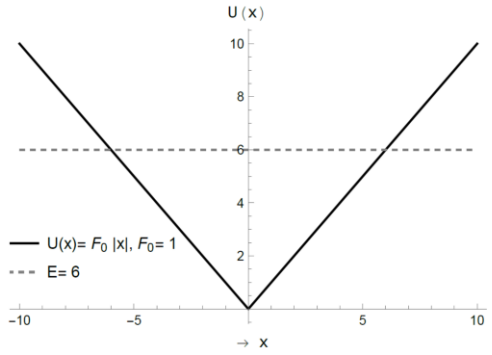
3. The turning points are the limits of the definite integral that represents the time period

$$T = \sqrt{2m}\int_{-x_0}^{x_0} \frac{dx}{\sqrt{E - U(x)}}$$

4. If the motion is bounded then find frequency of oscillations

$$k = \left(\frac{d^2U}{dx^2}\right)_{x=x_0} > 0 \quad \omega = \sqrt{\frac{k}{m}}$$

$$F(x) = \begin{cases} -F_0 & \text{for} \quad x > 0, \\ 0 & \text{for} \quad x = 0, \\ F_0 & \text{for} \quad x < 0. \end{cases}$$



$$\vec{F} = F(x)\hat{x} = \left(-B + \frac{A}{x^2}\right)\hat{x}$$

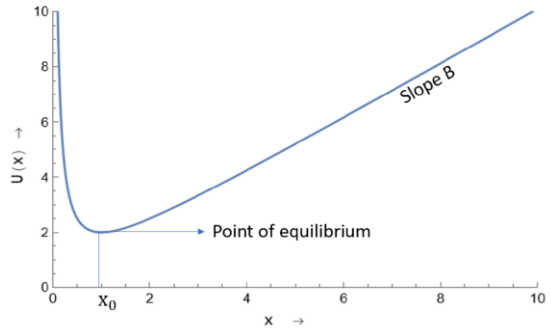


Figure 3: $U(x) = Bx + \frac{A}{x}$.

REVIEW: LINEAR 1D SYSTEMS

1. Find $U(x) \rightarrow$

$$U(x) = -\int_{x_0}^{x} F_x(x')dx'$$

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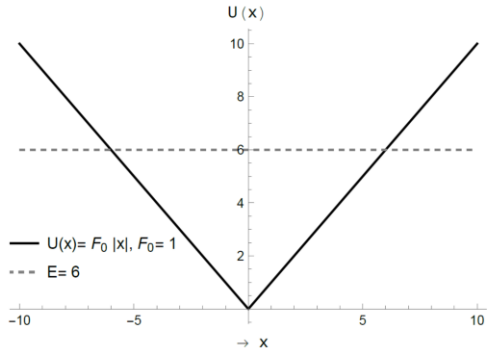
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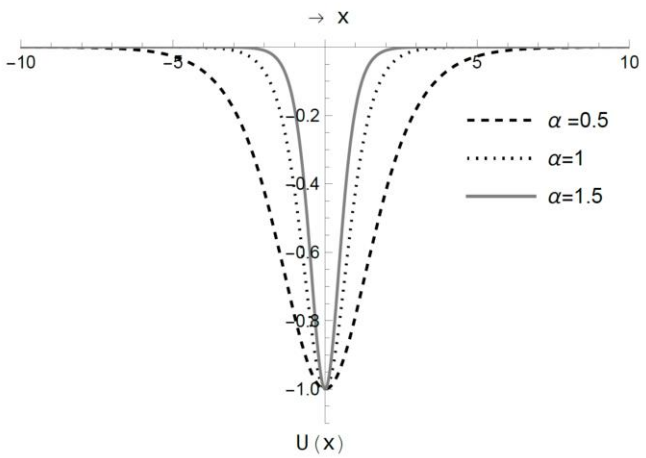
4. If the motion is bounded then find frequency of oscillations

$$k = \left(\frac{d^2U}{dx^2}\right)_{x=x_0} > 0 \quad \omega = \sqrt{\frac{k}{m}}$$

$$F(x) = \begin{cases} -F_0 & \text{for} \quad x > 0, \\ 0 & \text{for} \quad x = 0, \\ F_0 & \text{for} \quad x < 0. \end{cases}$$



$$U(x) = \frac{-4U_0}{(e^{\alpha x} + e^{-\alpha x})^2}$$



$$\vec{F} = F(x)\hat{x} = \left(-B + \frac{A}{x^2}\right)\hat{x}$$

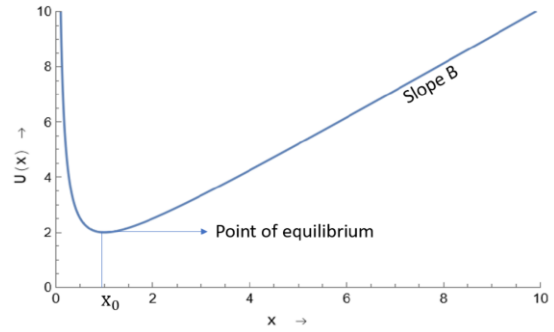


Figure 3: $U(x) = Bx + \frac{A}{x}$.

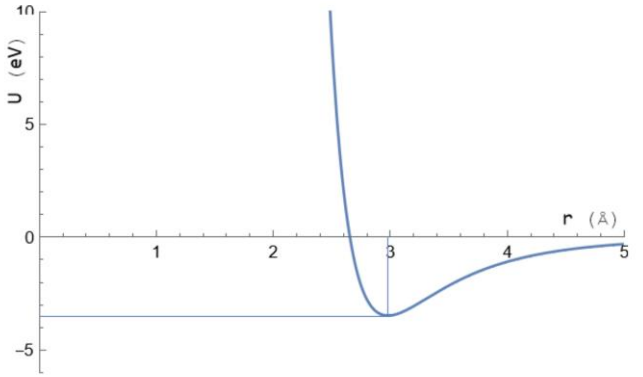


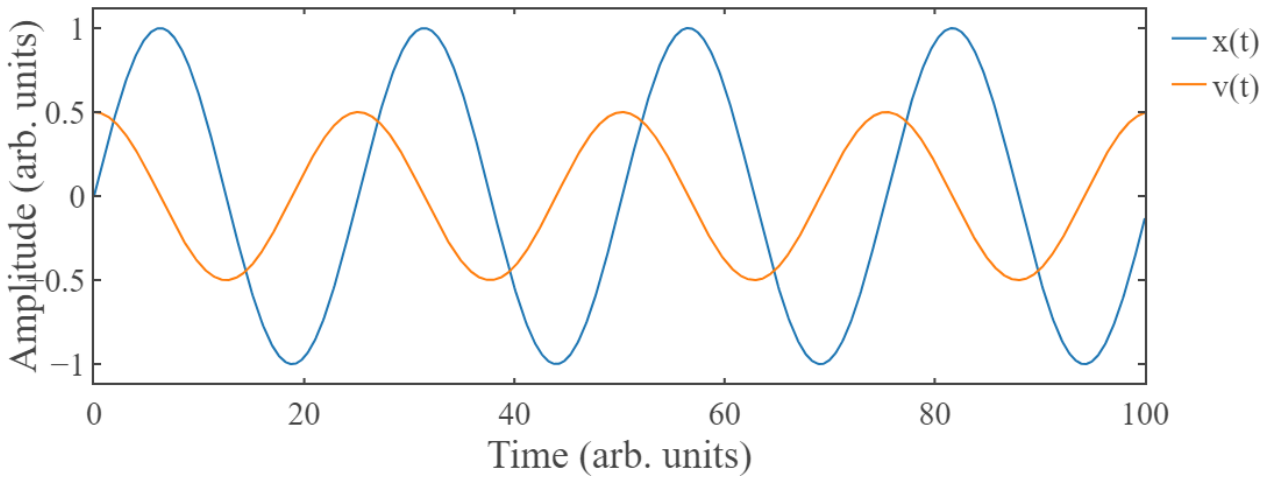
Figure 4: The Lennard-Jones potential $U(r) = \epsilon\left[\left(\frac{r_0}{r}\right)^{12} - 2\left(\frac{r_0}{r}\right)^{6}\right]$.

REVIEW: PHASE PORTRAITS

Linear harmonic oscillator $U(x) = Kx^2/2$ with $K > 0$

Each phase trajectory is an ellipse given by

$$\frac{1}{2}m\dot{x}^2 + U(x) = E \quad \Rightarrow \quad \boxed{\frac{x^2}{(2E/K)} + \frac{\dot{x}^2}{(2E/m)} = 1}$$



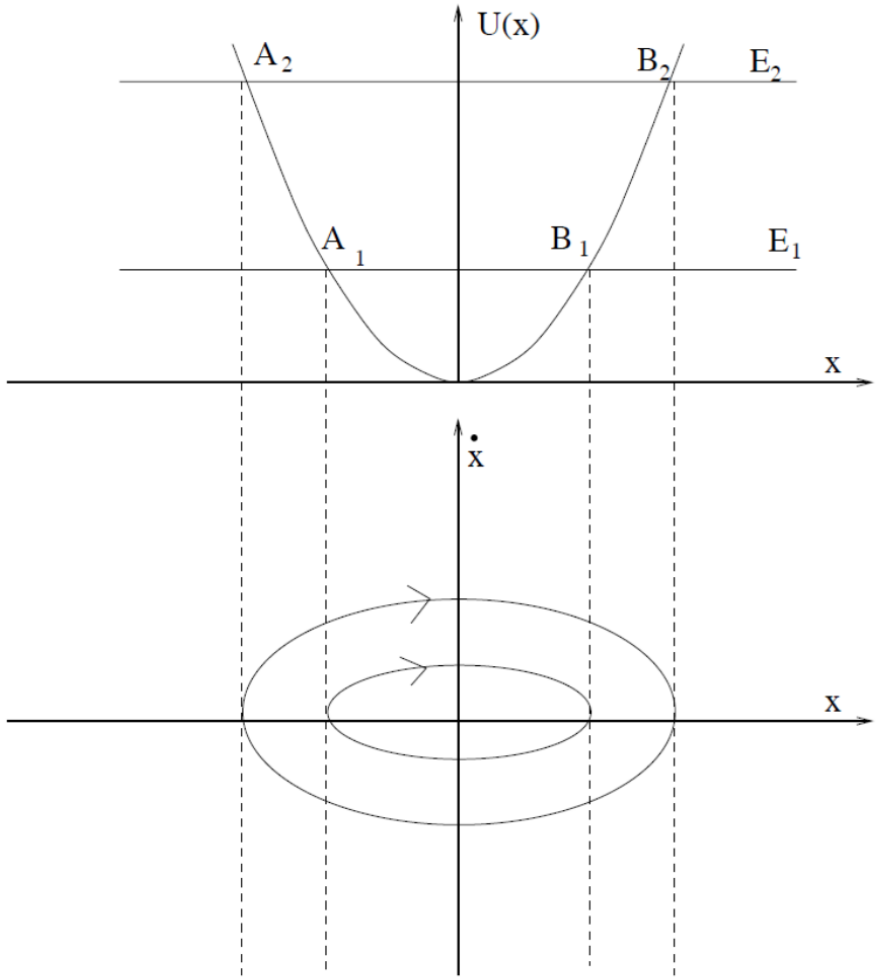


Figure 5.2: Phase portrait for $U(x) = Kx^2/2$

REVIEW: DAMPED OSCILLATIONS

Undamped β is zero

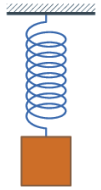
$$x(t) = C_1 e^{i\omega_o t} + C_2 e^{-i\omega_o t}$$

Weak Damping $\beta < \omega_o$

$$x(t) = A e^{-\beta t} \cos(\omega_1 t - \delta)$$

Strong Damping $\beta > \omega_o$

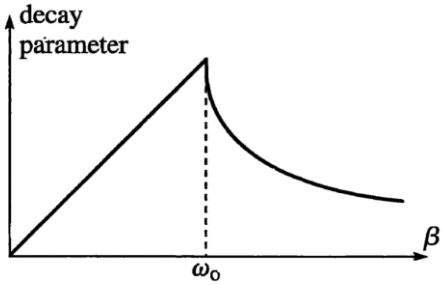
$$x(t) = C_1 e^{-\left(\beta - \sqrt{\beta^2 - \omega_o^2}\right)t} + C_2 e^{-\left(\beta + \sqrt{\beta^2 - \omega_o^2}\right)t}$$

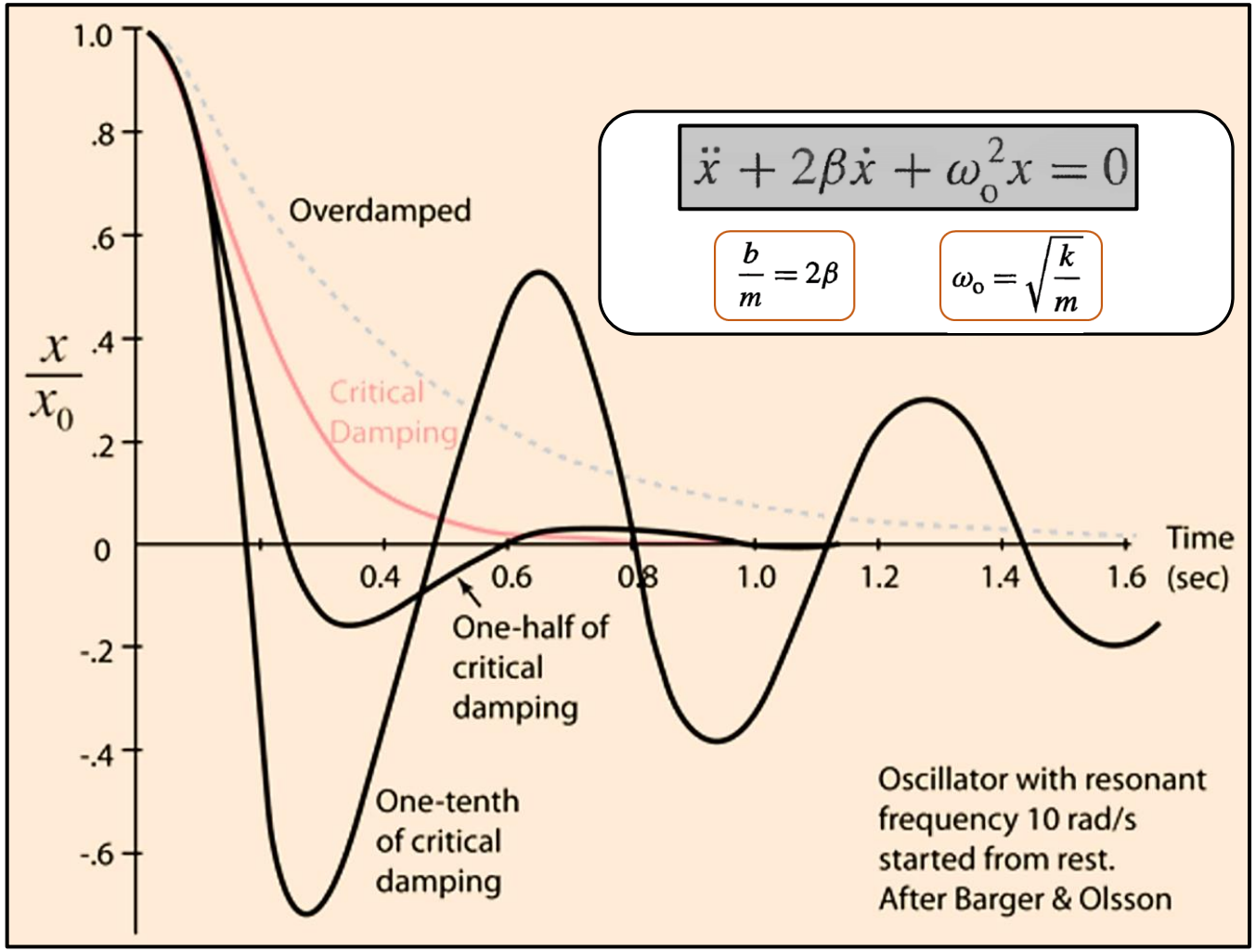


Critical Damping $\beta = \omega_o$

$$x(t) = C_1 e^{-\beta t} + C_2 t e^{-\beta t}$$

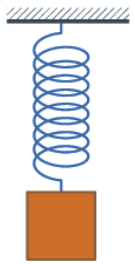
damping	β	decay parameter
none	$\beta = 0$	0
under	$\beta < \omega_o$	β
critical	$\beta = \omega_o$	β
over	$\beta > \omega_o$	$\beta - \sqrt{\beta^2 - \omega_o^2}$





LINEAR 1D SYSTEMS: EXAMPLE

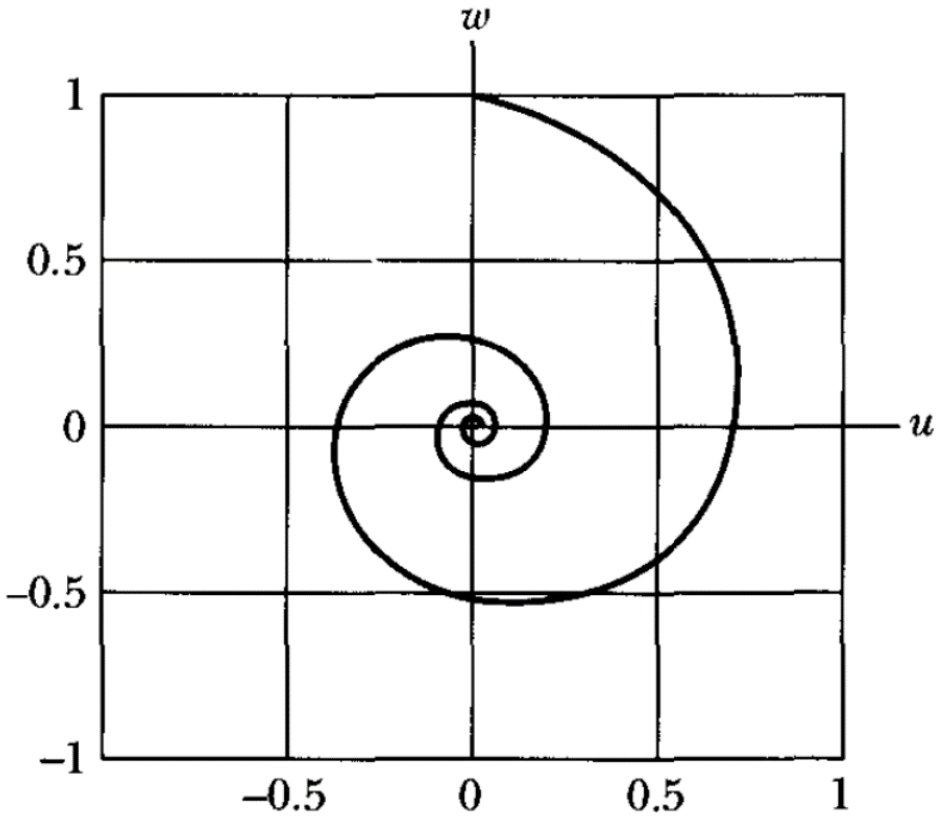
Underdamped oscillator



Weak Damping $\beta < \omega_o$

$$x(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta)$$

$$\omega_1 = \sqrt{\omega_o^2 - \beta^2} = \text{frequency of damped oscillator}$$



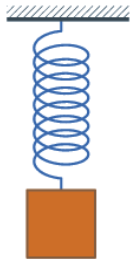
$$u = \omega_1 x, \quad w = \beta x + \dot{x}$$

$$u = \omega_1 A e^{-\beta t}\cos(\omega_1 t - \delta)$$

$$w = -\omega_1 A e^{-\beta t}\sin(\omega_1 t - \delta)$$

LINEAR 1D SYSTEMS: EXAMPLE

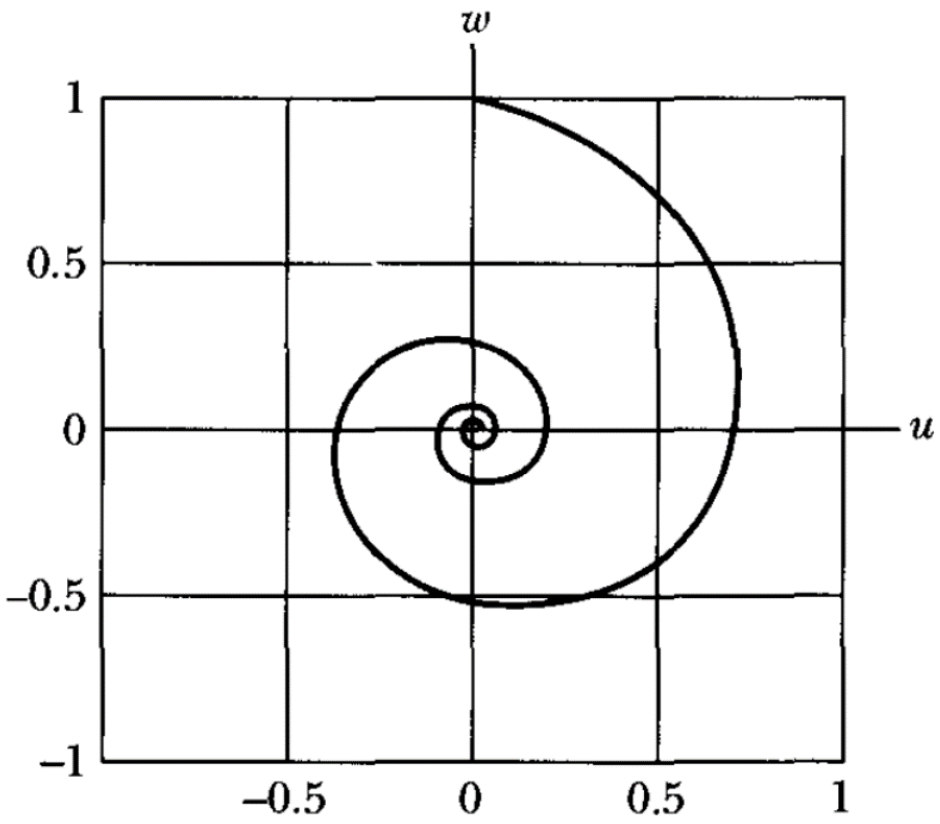
Underdamped oscillator

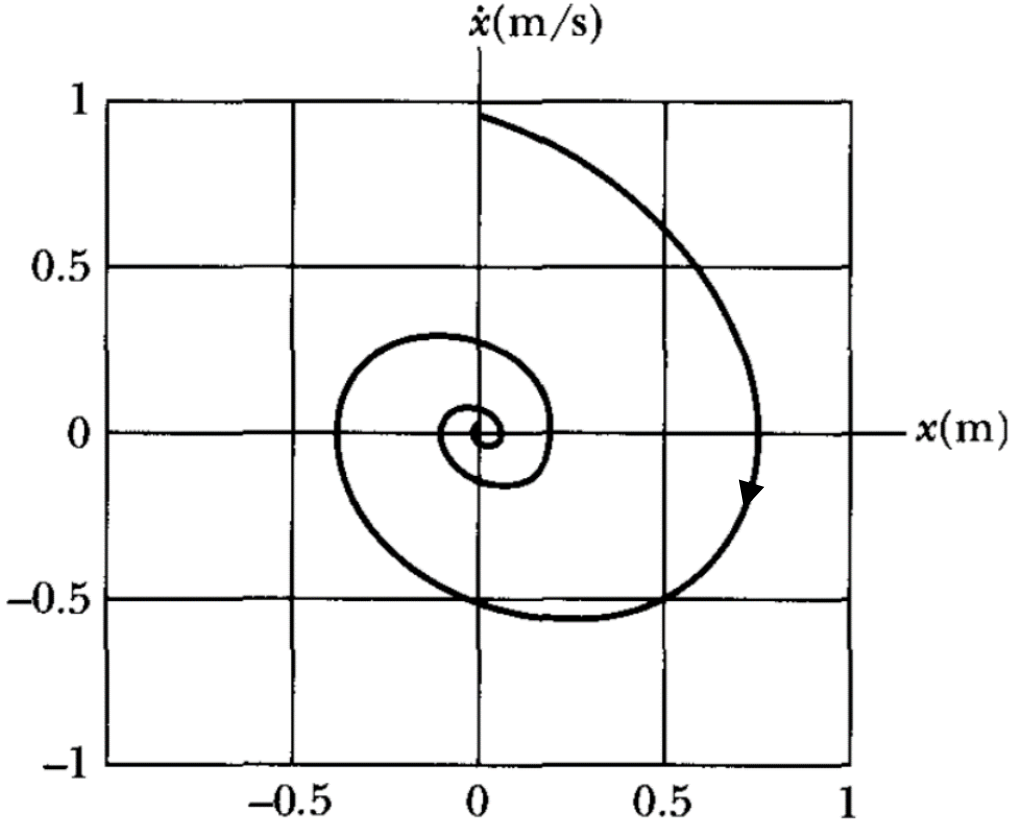


Weak Damping $\beta < \omega_o$

$$x(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta)$$

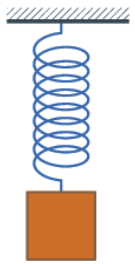
$$\omega_1 = \sqrt{\omega_o^2 - \beta^2} = \text{frequency of damped oscillator}$$





LINEAR 1D SYSTEMS: EXAMPLE

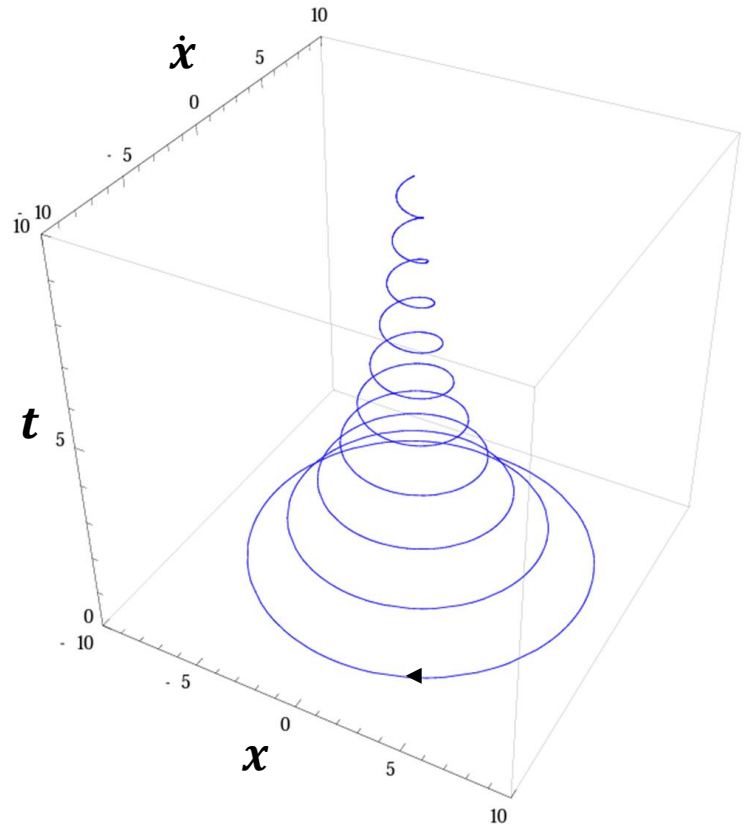
Underdamped oscillator



Weak Damping $\beta < \omega_o$

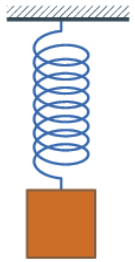
$$x(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta)$$

$$\omega_1 = \sqrt{\omega_o^2 - \beta^2} = \text{frequency of damped oscillator}$$



LINEAR 1D SYSTEMS: EXAMPLE

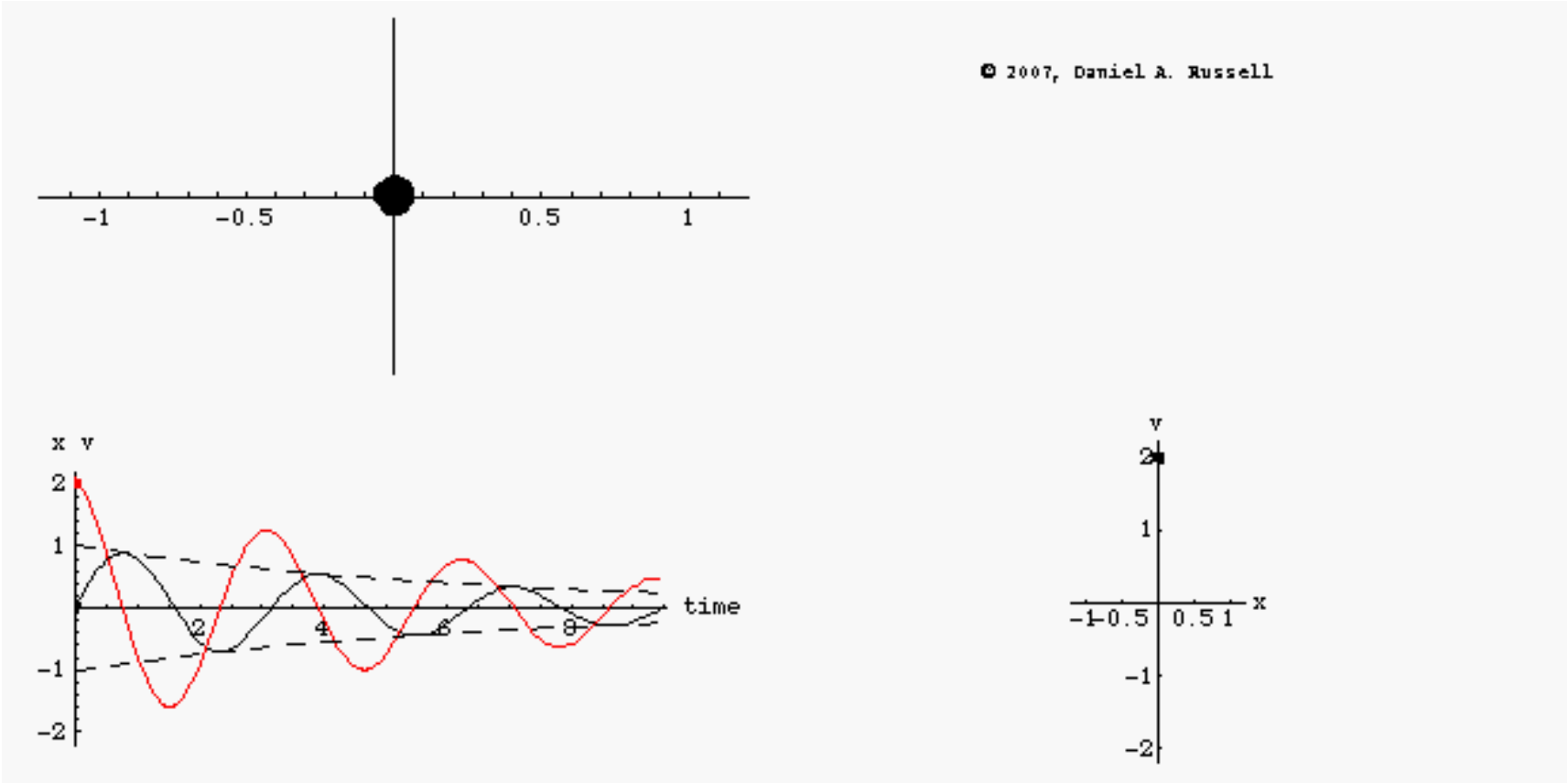
Underdamped oscillator



Weak Damping $\beta < \omega_o$

$$x(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta)$$

$$\omega_1 = \sqrt{\omega_o^2 - \beta^2} = \text{frequency of damped oscillator}$$



REVIEW: PHASE PORTRAITS

Overdamped oscillator

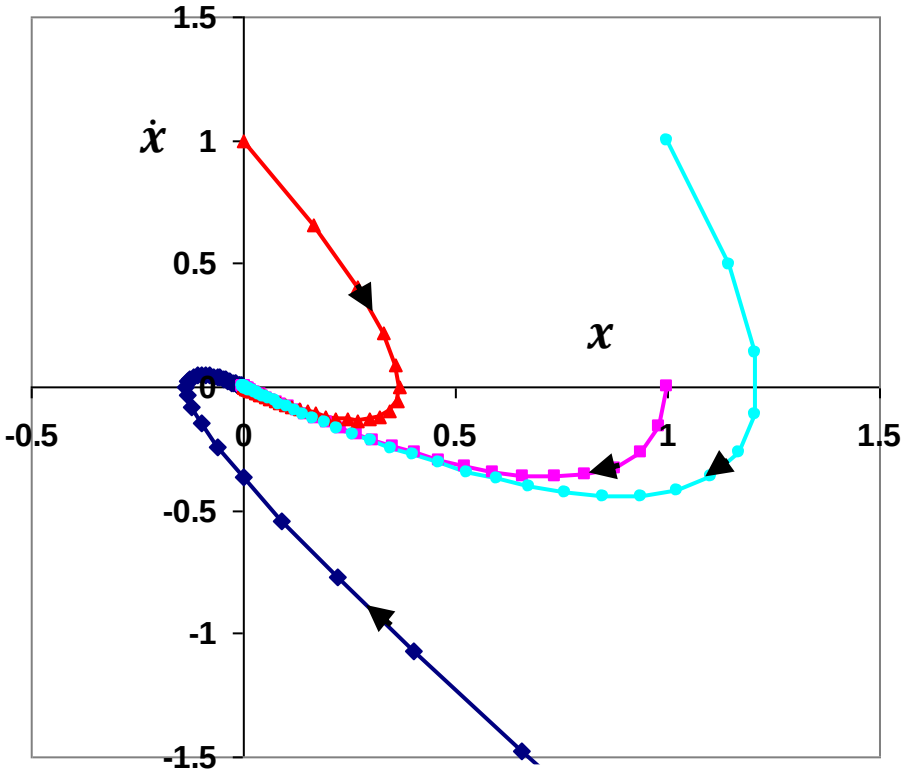


Strong Damping $\beta > \omega_o$

$$x(t) = C_1 e^{-\left(\beta - \sqrt{\beta^2 - \omega_o^2}\right)t} + C_2 e^{-\left(\beta + \sqrt{\beta^2 - \omega_o^2}\right)t}$$

$$\dot{x} = -(\beta - \omega_2)x$$

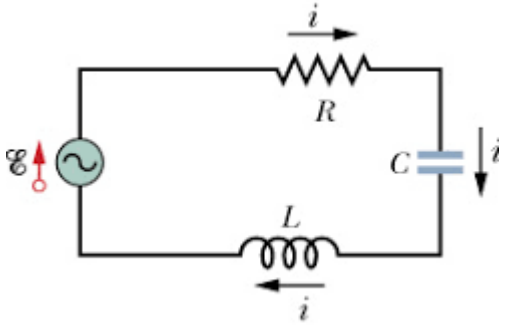
$$\omega_2 = \sqrt{\beta^2 - \omega_0^2}$$



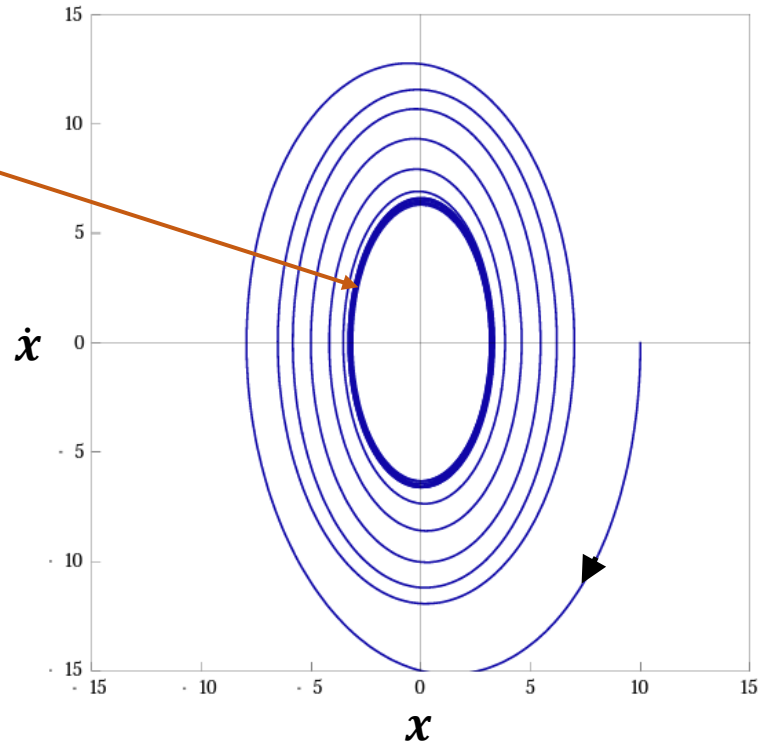
BOUNDED MOTION: EXAMPLES

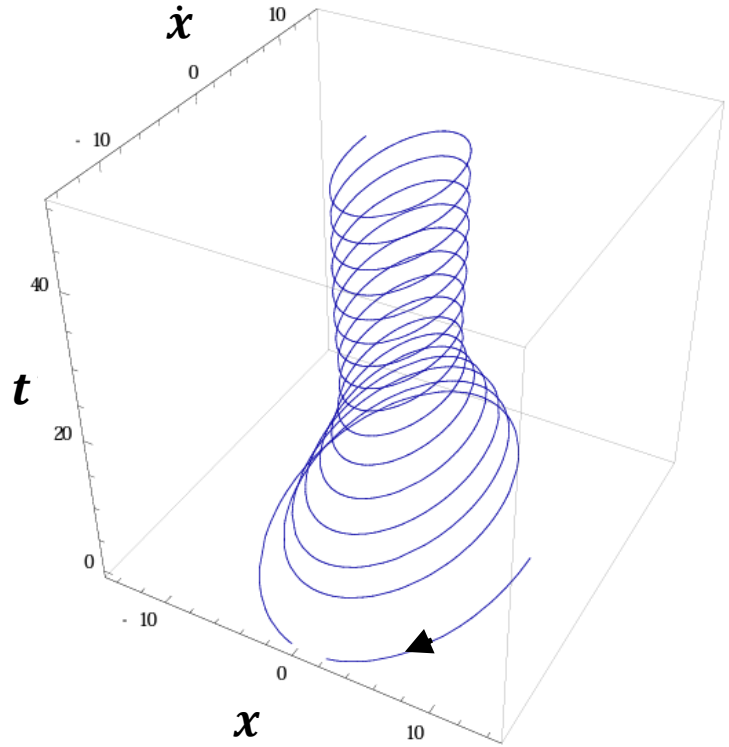
Driven underdamped oscillator

$$x(t) = A\cos(\omega t - \delta) + A_{\text{tr}} e^{-\beta t}\cos(\omega_1 t - \delta_{\text{tr}})$$



Limit cycle (attractor)

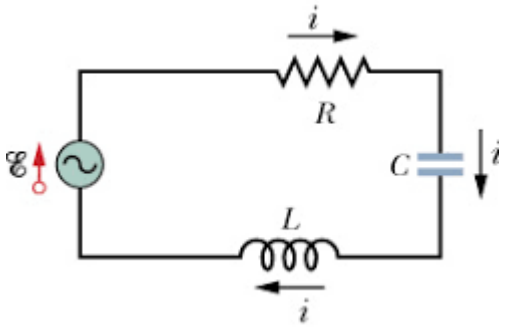




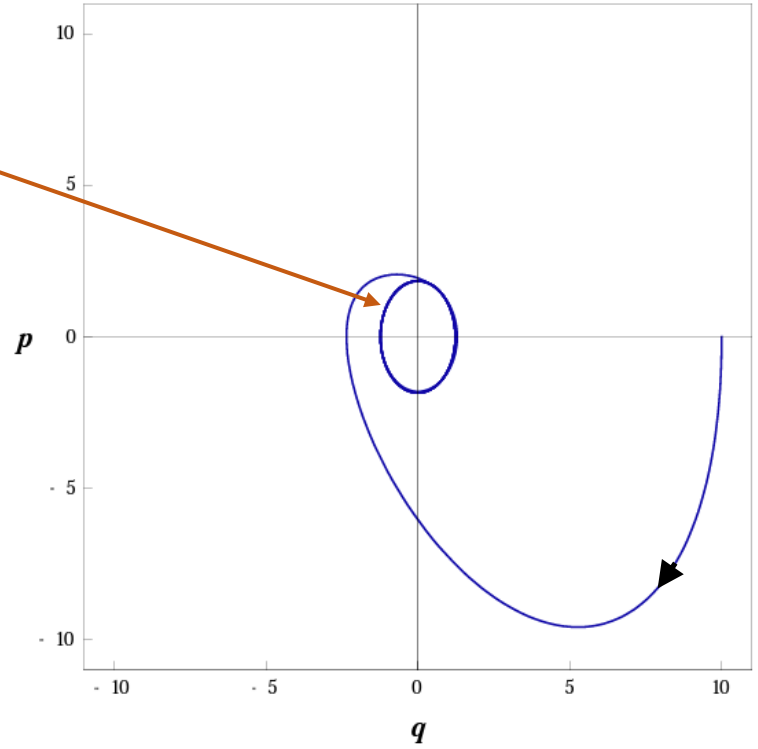
BOUNDED MOTION: EXAMPLES

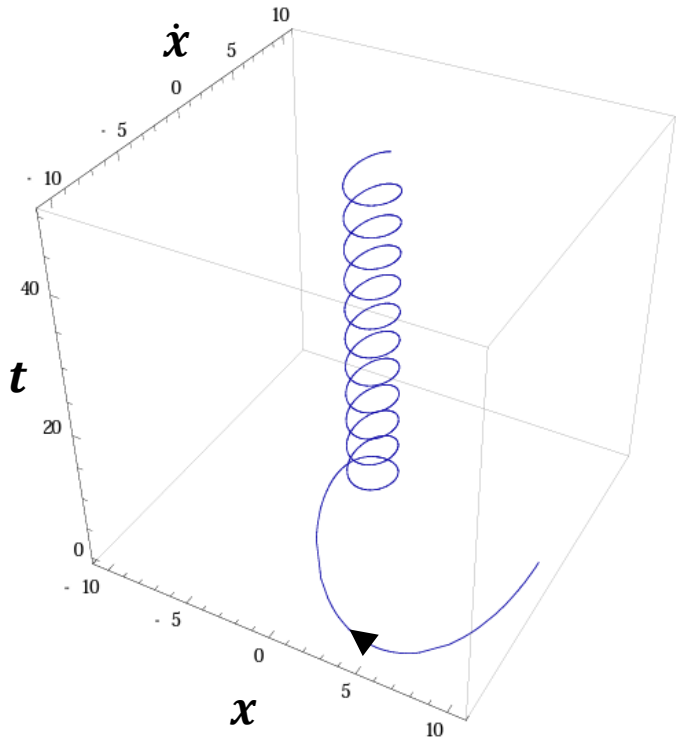
Driven overdamped oscillator

$$x(t) = A\cos(\omega t - \delta) + A_{\text{tr}} e^{-\beta t}\cos(\omega_1 t - \delta_{\text{tr}})$$



Limit cycle (attractor)

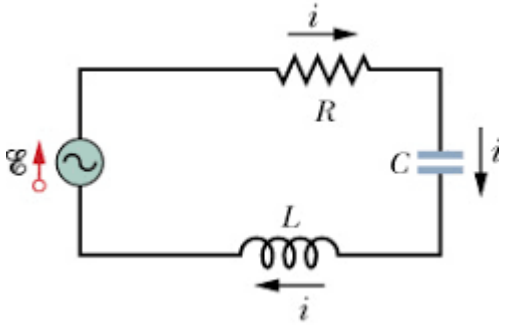




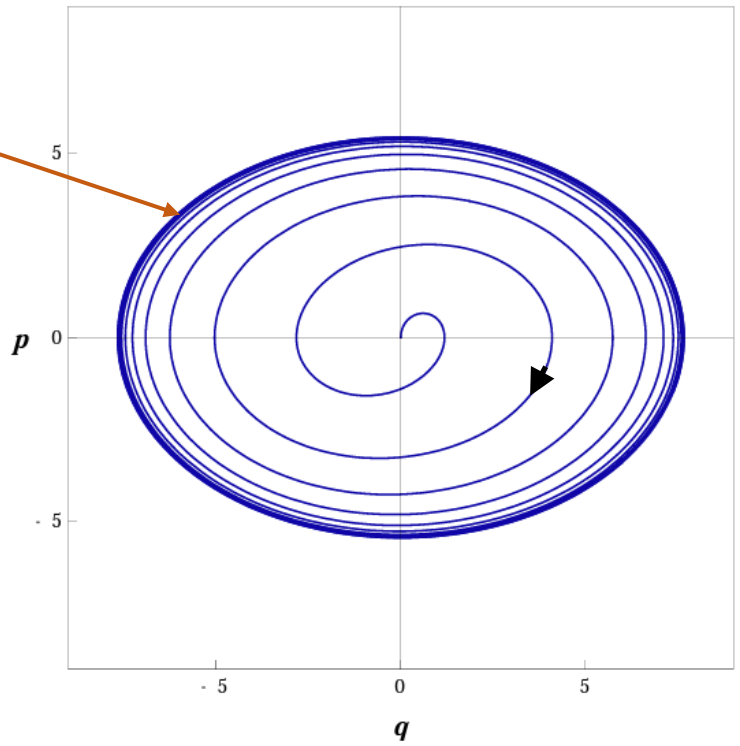
BOUNDED MOTION: EXAMPLES

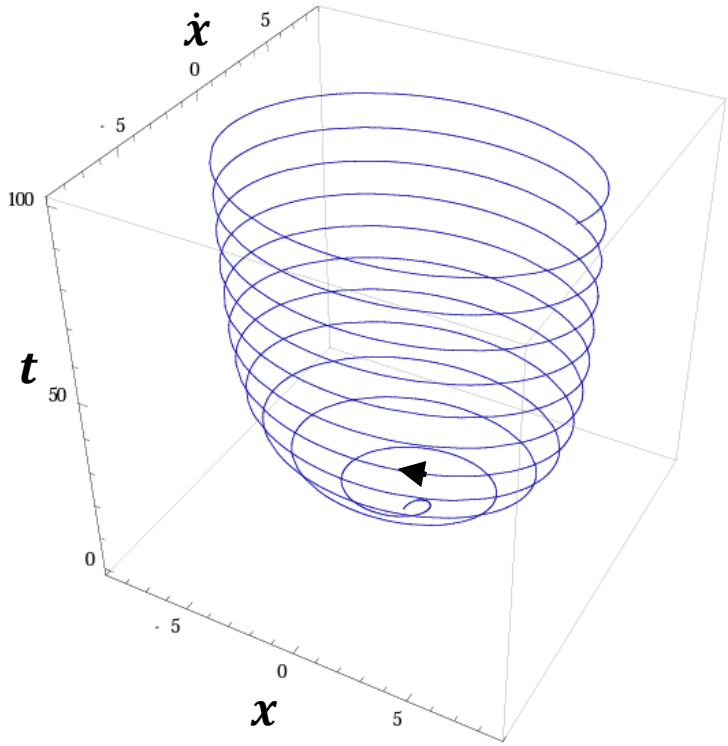
Driven damped oscillator, at resonance

$$x(t) = A\cos(\omega t - \delta) + A_{\mathrm{tr}} e^{-\beta t}\cos(\omega_1 t - \delta_{\mathrm{tr}})$$



Limit cycle (attractor)

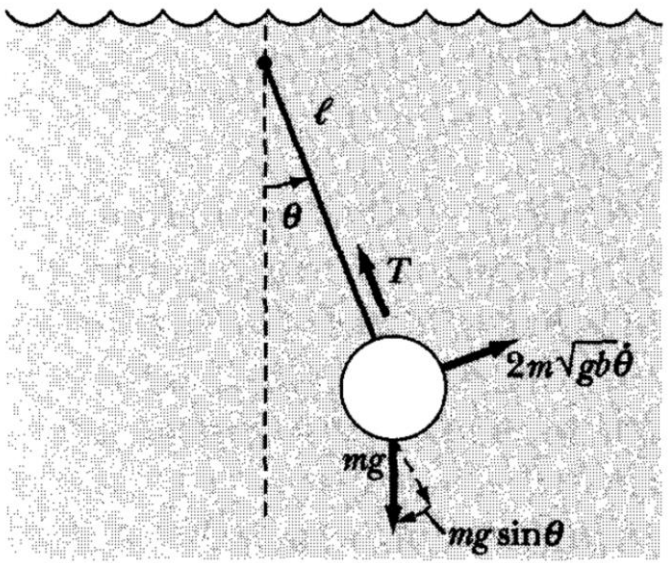




LINEAR 1D SYSTEMS: EXAMPLE

EXAMPLE 3.3

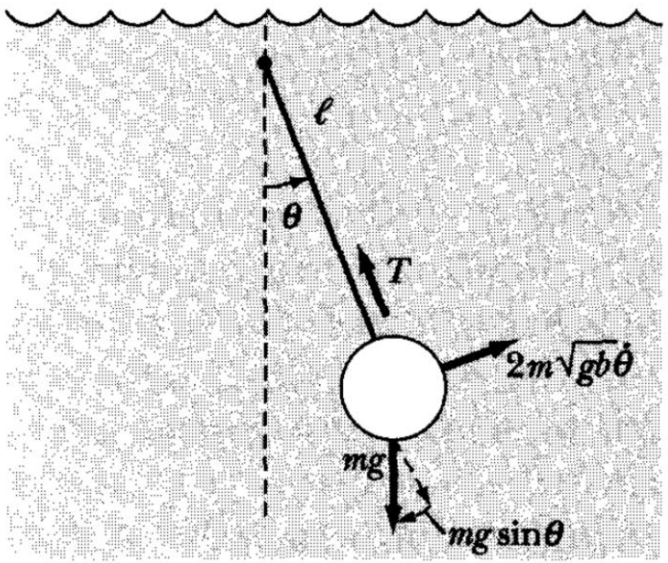
Consider a pendulum of length ℓ and a bob of mass m at its end (Figure 3-13) moving through oil with θ decreasing. The massive bob undergoes small oscillations, but the oil retards the bob's motion with a resistive force proportional to the speed with $F_{\text{res}} = 2m\sqrt{g/\ell}\ (\ell\dot{\theta})$. The bob is initially pulled back at $t = 0$ with $\theta = \alpha$ and $\dot{\theta} = 0$. Find the angular displacement θ and velocity $\dot{\theta}$ as a function of time. Sketch the phase diagram if $\sqrt{g/\ell} = 10\ \text{s}^{-1}$ and $\alpha = 10^{-2}$ rad.



LINEAR 1D SYSTEMS: EXAMPLE

EXAMPLE 3.3

Consider a pendulum of length ℓ and a bob of mass m at its end (Figure 3-13) moving through oil with θ decreasing. The massive bob undergoes small oscillations, but the oil retards the bob's motion with a resistive force proportional to the speed with $F_{\text{res}} = 2m\sqrt{g/\ell}\ (\ell\dot{\theta})$. The bob is initially pulled back at $t = 0$ with $\theta = \alpha$ and $\dot{\theta} = 0$. Find the angular displacement θ and velocity $\dot{\theta}$ as a function of time. Sketch the phase diagram if $\sqrt{g/\ell} = 10\ \text{s}^{-1}$ and $\alpha = 10^{-2}$ rad.



NEWTON'S LAWS: POLAR COORDINATES

Solution

$$\mathbf{a} = \left(\ddot{r} - r\dot{\phi}^2\right)\hat{\mathbf{r}} + \left(r\ddot{\phi} + 2\dot{r}\dot{\phi}\right)\hat{\boldsymbol{\phi}}$$

$$\mathbf{F} = m\mathbf{a} \iff \begin{cases} F_r = m(\ddot{r} - r\dot{\phi}^2) \\ F_\phi = m(r\ddot{\phi} + 2\dot{r}\dot{\phi}) \end{cases}$$

For constant r

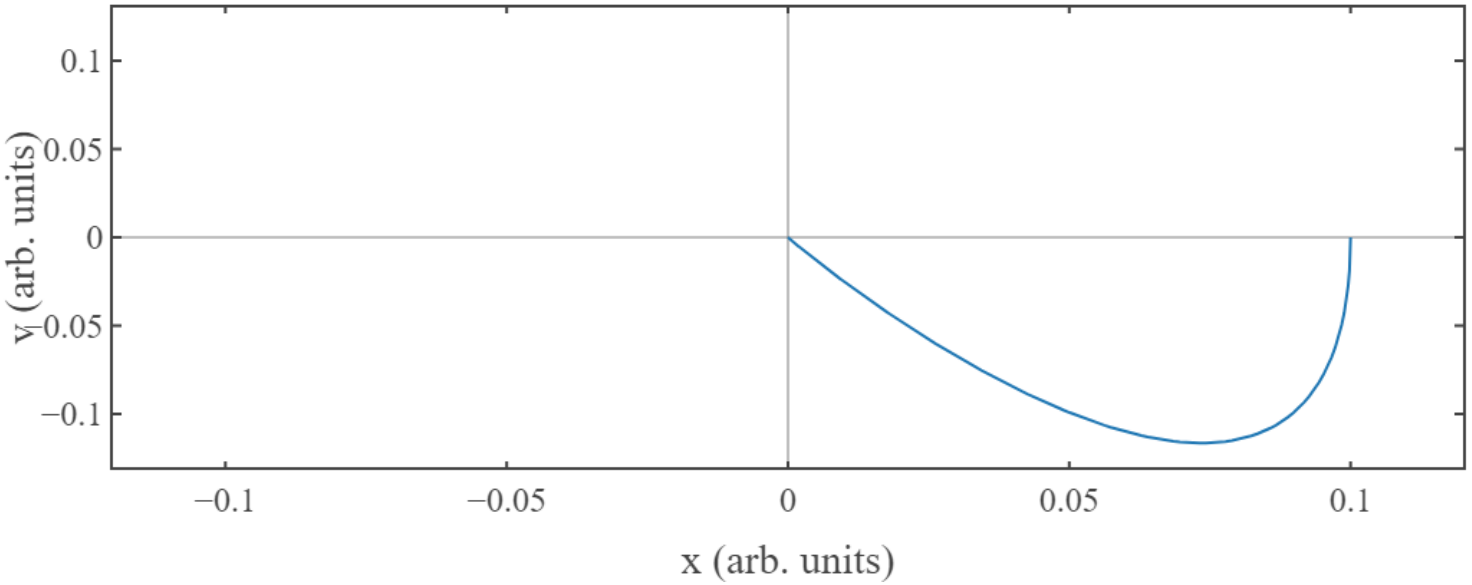
$$\mathbf{a} = -r\dot{\phi}^2\hat{\mathbf{r}} + r\ddot{\phi}\hat{\boldsymbol{\phi}} \qquad \mathbf{a} = -r\omega^2\hat{\mathbf{r}} + r\alpha\hat{\boldsymbol{\phi}},$$

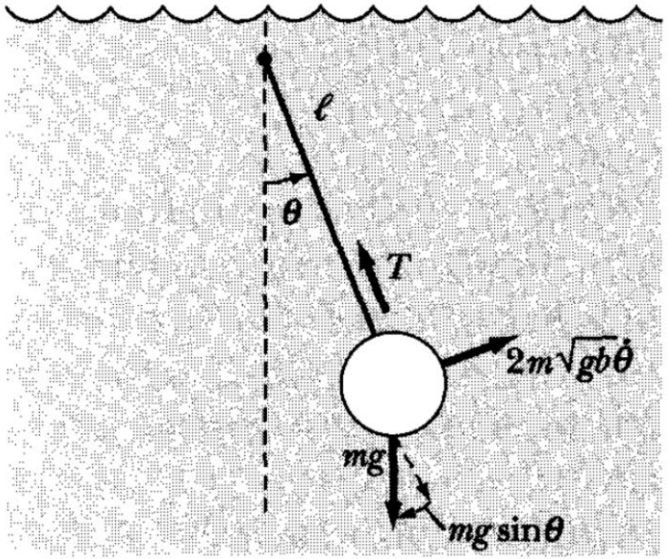
LINEAR 1D SYSTEMS: EXAMPLE

Critically damped oscillator

$$\theta(t) = \alpha(1 + \sqrt{g/\ell}\, t)e^{-\sqrt{g/\ell}\, t}$$

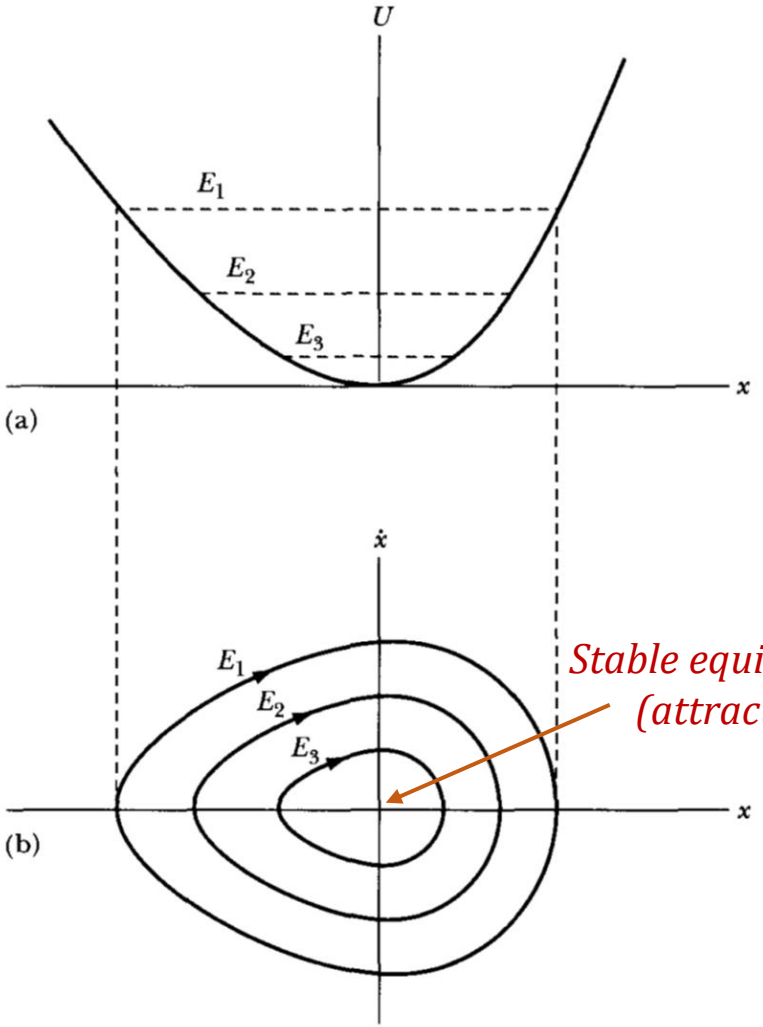
$$\dot{\theta}(t) = \frac{-\alpha g}{\ell} t e^{-\sqrt{g/\ell}\, t}$$





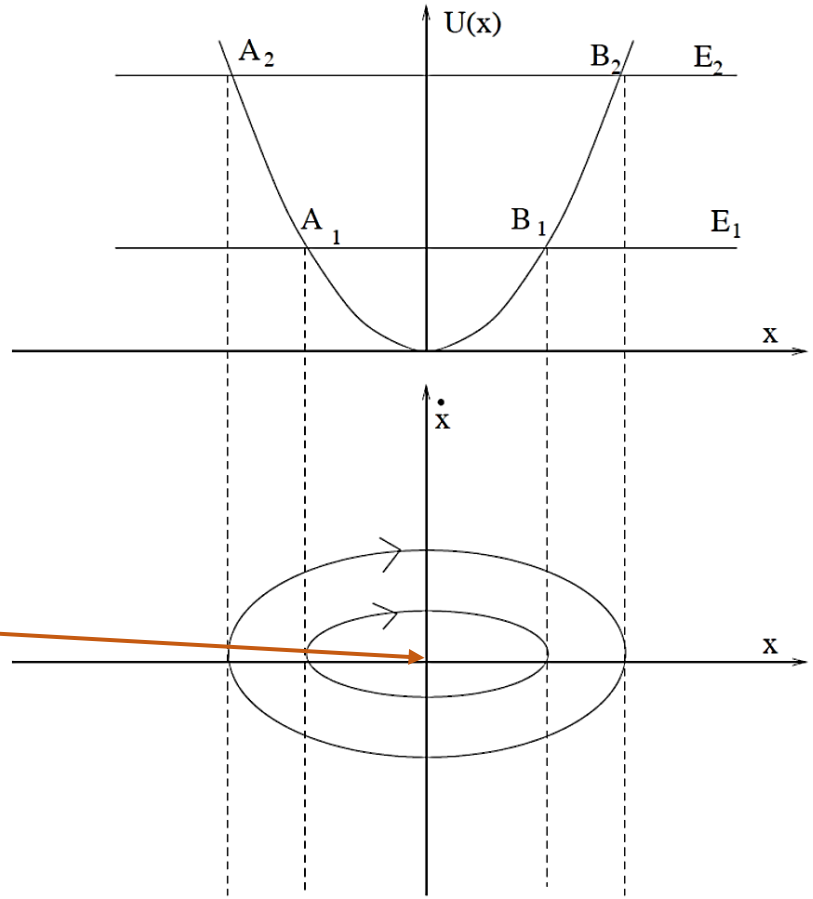
REVIEW: PHASE PORTRAITS

Example: Asymmetric potential



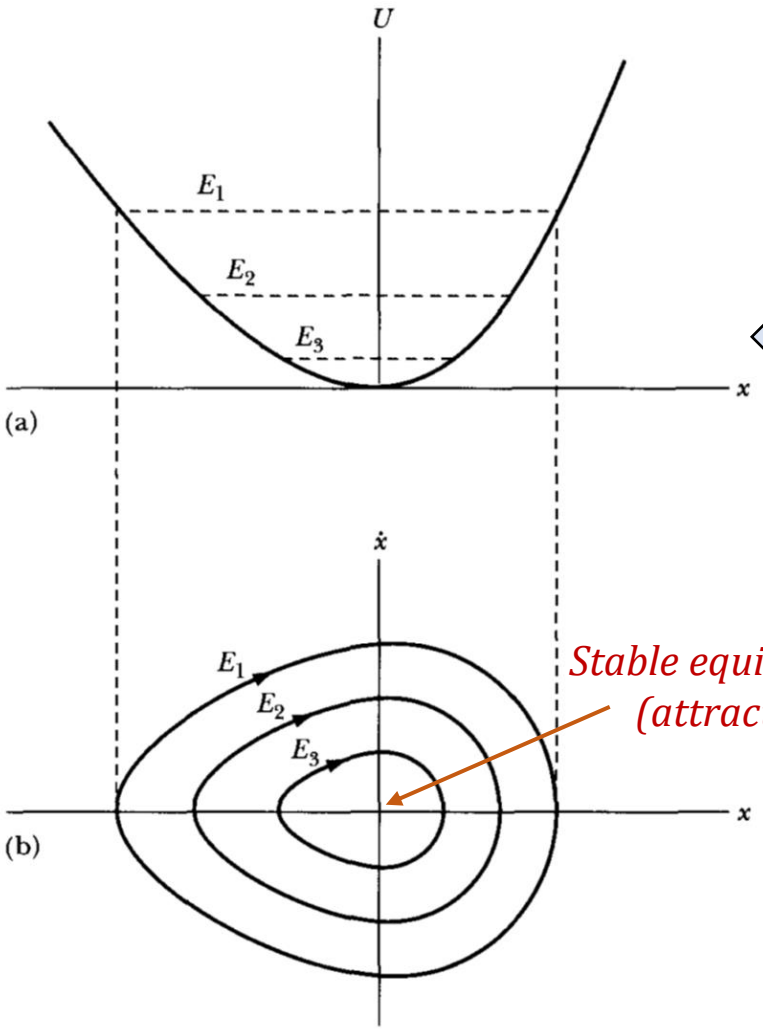
Stable equilibrium (attractor)

Example: Symmetric potential



REVIEW: PHASE PORTRAITS

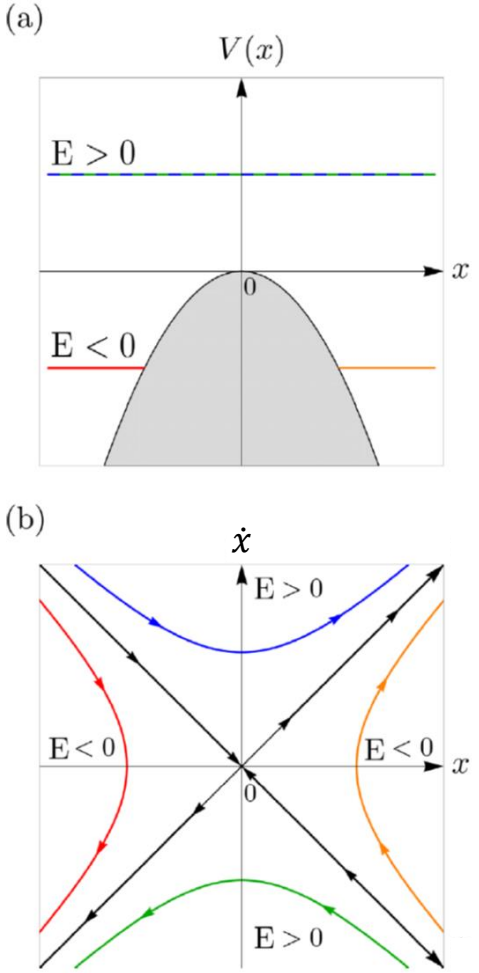
Example: Asymmetric potential



Stable equilibrium (attractor)

Bounded & Unbounded

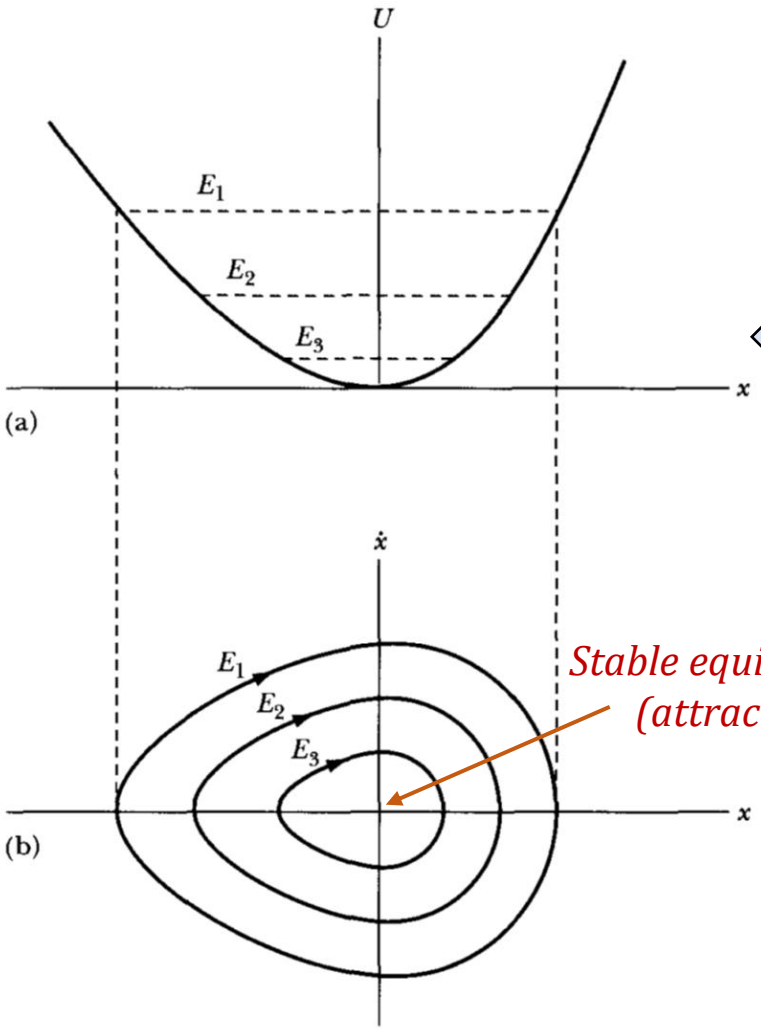
Example: Inverted symmetric



AVS Quantum Sci.4, 024402 (2022)

REVIEW: PHASE PORTRAITS

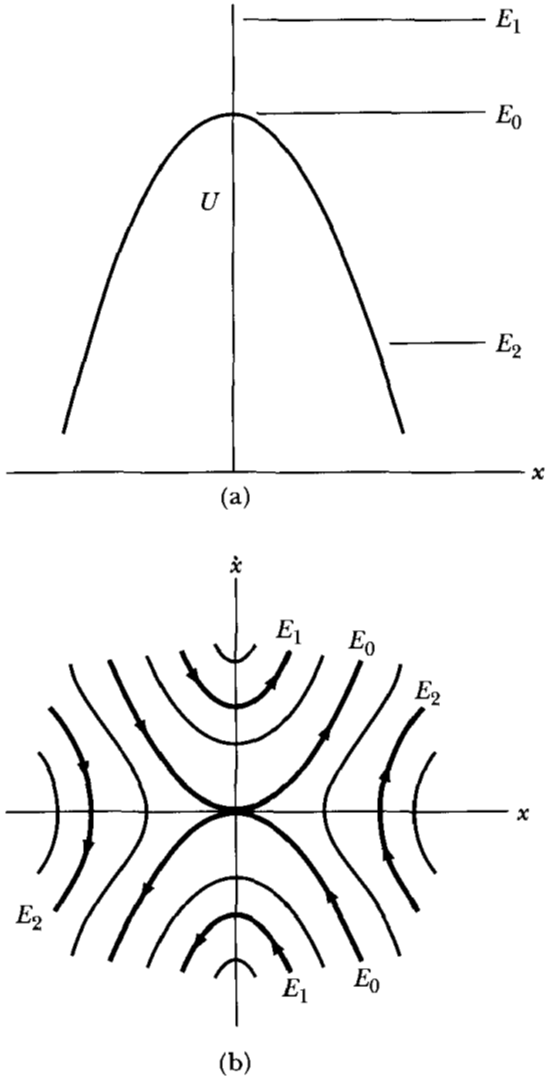
Example: Asymmetric potential



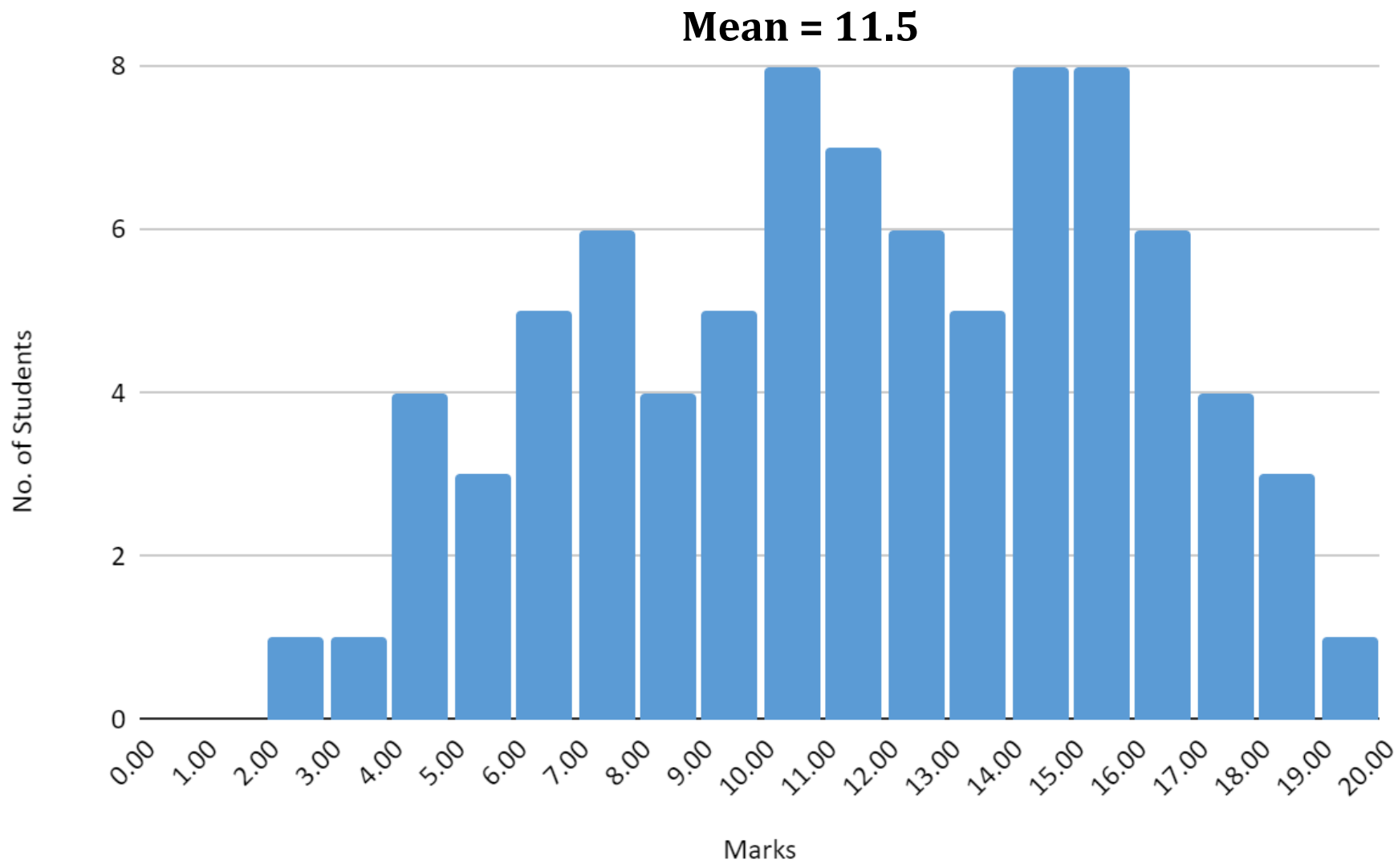
Stable equilibrium (attractor)

Bounded & Unbounded

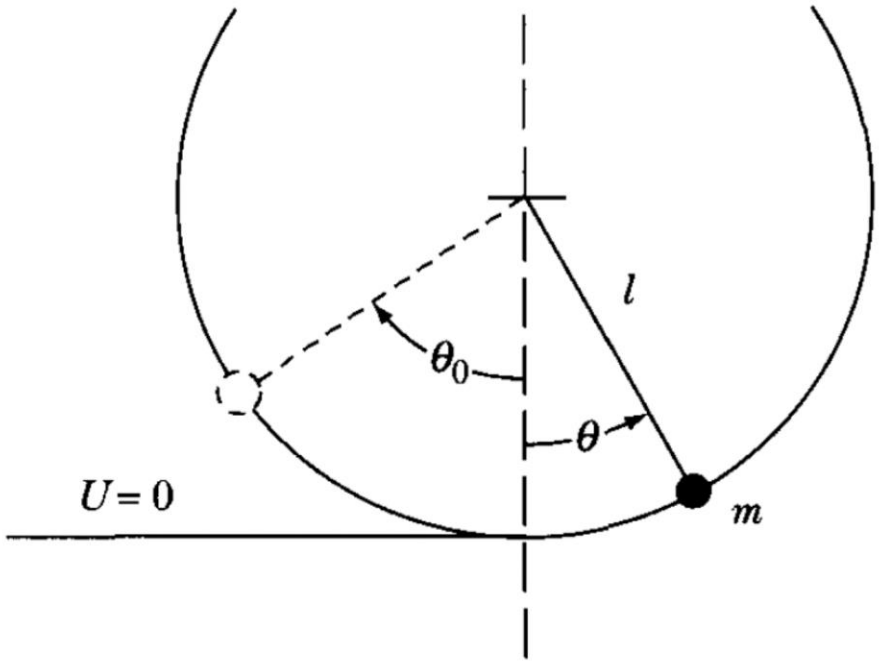
Example: Inverted asymmetric



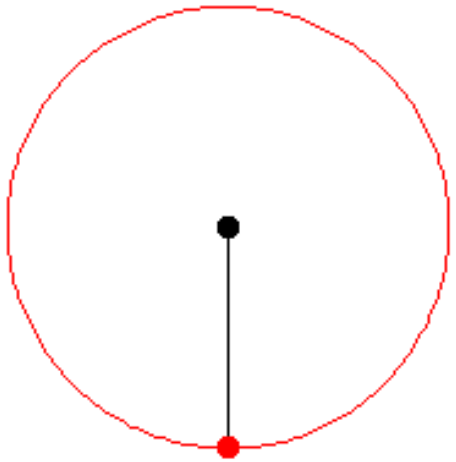
QUIZ 1: MARK DISTRIBUTION

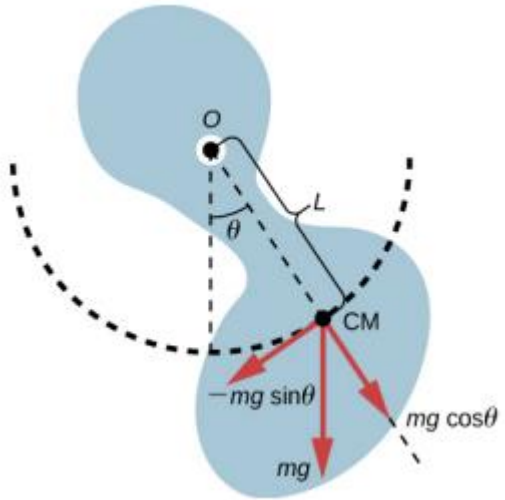


NONLINEAR SYSTEM: PLANE PENDULUM

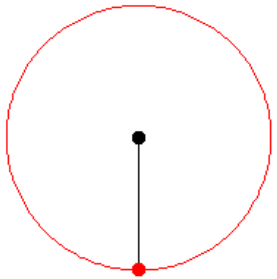


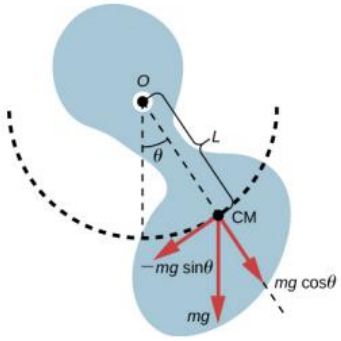
The plane pendulum where the mass m is not required to oscillate in small angles. The angle $\theta > 0$ is in the counterclockwise direction so that $\theta_0 < 0$.



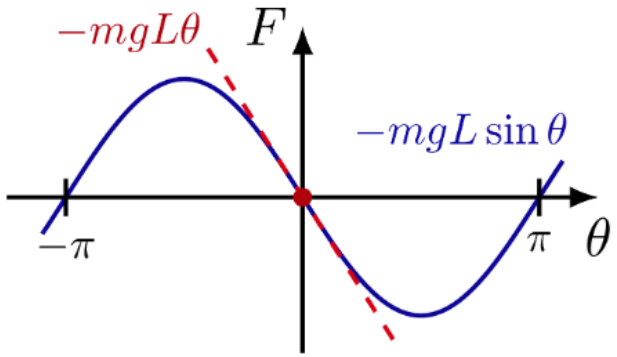


NONLINEAR SYSTEM: PLANE PENDULUM





Approximate the gravitational force on a pendulum with a Hooke's law, which is equivalent to the small-angle approximation:

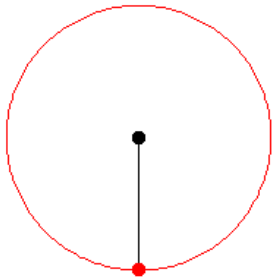


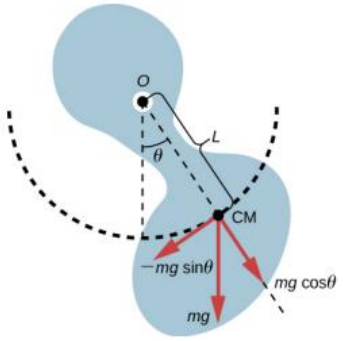
Small angle approximation

$$\ddot{\theta} + \omega_0^2 \theta = 0$$

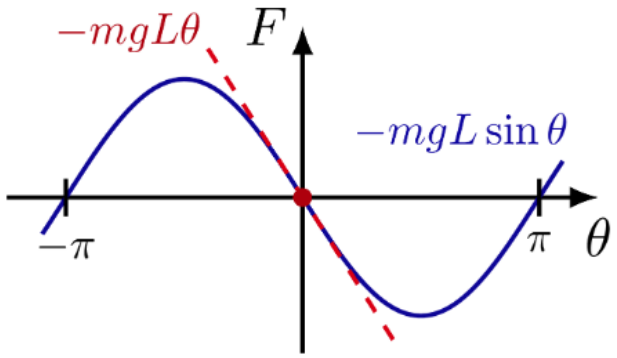
$$\tau \cong 2\pi\sqrt{\frac{l}{g}}$$

NONLINEAR SYSTEM: PLANE PENDULUM





Approximate the gravitational force on a pendulum with a Hooke's law, which is equivalent to the small-angle approximation:



For a parabolic potential

$$\frac{1}{2}m\,\dot{x}^2 + U(x) = E$$

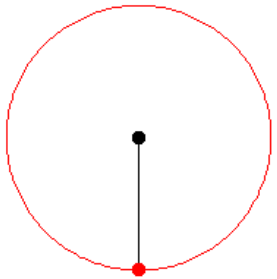
⇩

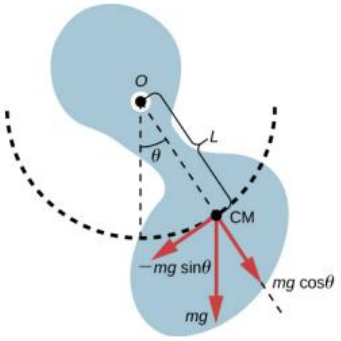
$$\dot{x}(x) = \pm\sqrt{\frac{2}{m}}\sqrt{E - U(x)}$$

⇩

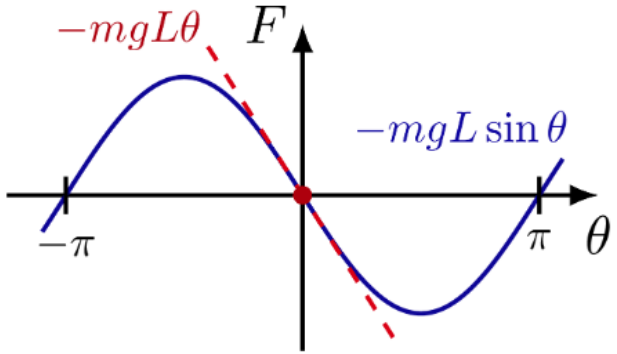
$$t = \int_{x_o}^{x} \frac{dx'}{\dot{x}(x')} = \sqrt{\frac{m}{2}}\int_{x_o}^{x} \frac{dx'}{\sqrt{E - U(x')}}$$

NONLINEAR SYSTEM: PLANE PENDULUM

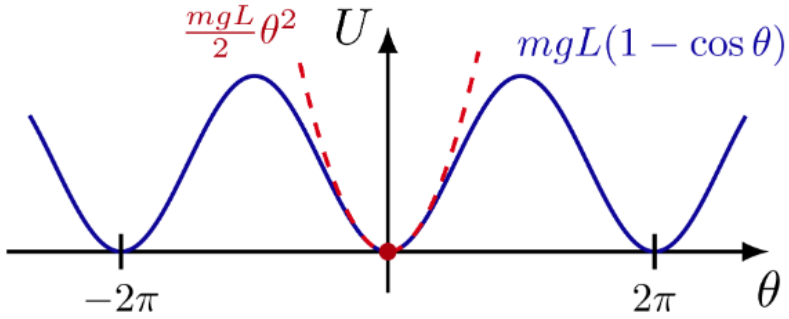




Approximate the gravitational force on a pendulum with a Hooke's law, which is equivalent to the small-angle approximation:



Approximate the potential energy of a pendulum with a parabola to get a simple harmonic oscillator:



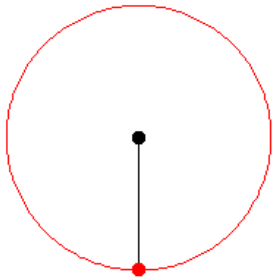
For a parabolic potential

$$\frac{1}{2}m\,\dot{x}^2 + U(x) = E$$

$$\dot{x}(x) = \pm\sqrt{\frac{2}{m}}\sqrt{E - U(x)}$$

$$t = \int_{x_0}^{x} \frac{dx'}{\dot{x}(x')} = \sqrt{\frac{m}{2}}\int_{x_0}^{x} \frac{dx'}{\sqrt{E - U(x')}}$$

NONLINEAR SYSTEM: PLANE PENDULUM



Thornton-Marion Appendix B.2

Transform the elliptic integral

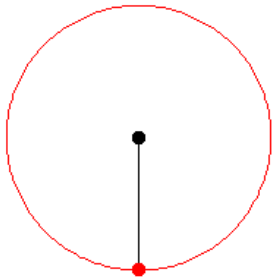
$$\int_0^{\phi} \frac{d\theta}{\sqrt{1 - n^2 \sin^2 \theta}}, \quad \text{where } n^2 > 1$$

into a standard form.

Solution. To reduce this integral to standard form, the radical must be transformed to $\sqrt{1 - k^2 \sin^2 \theta}$, with $k^2 < 1$. To do this, consider the transformation $n \sin \theta = \sin \beta$.

$$\tau \cong 2\pi \sqrt{\frac{l}{g}} \left[1 + \frac{1}{16} \theta_0^2 + \frac{11}{3072} \theta_0^4 \right]$$

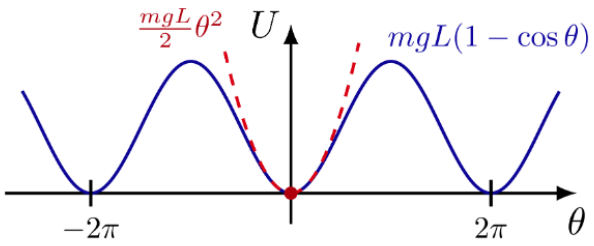
NONLINEAR SYSTEM: PLANE PENDULUM

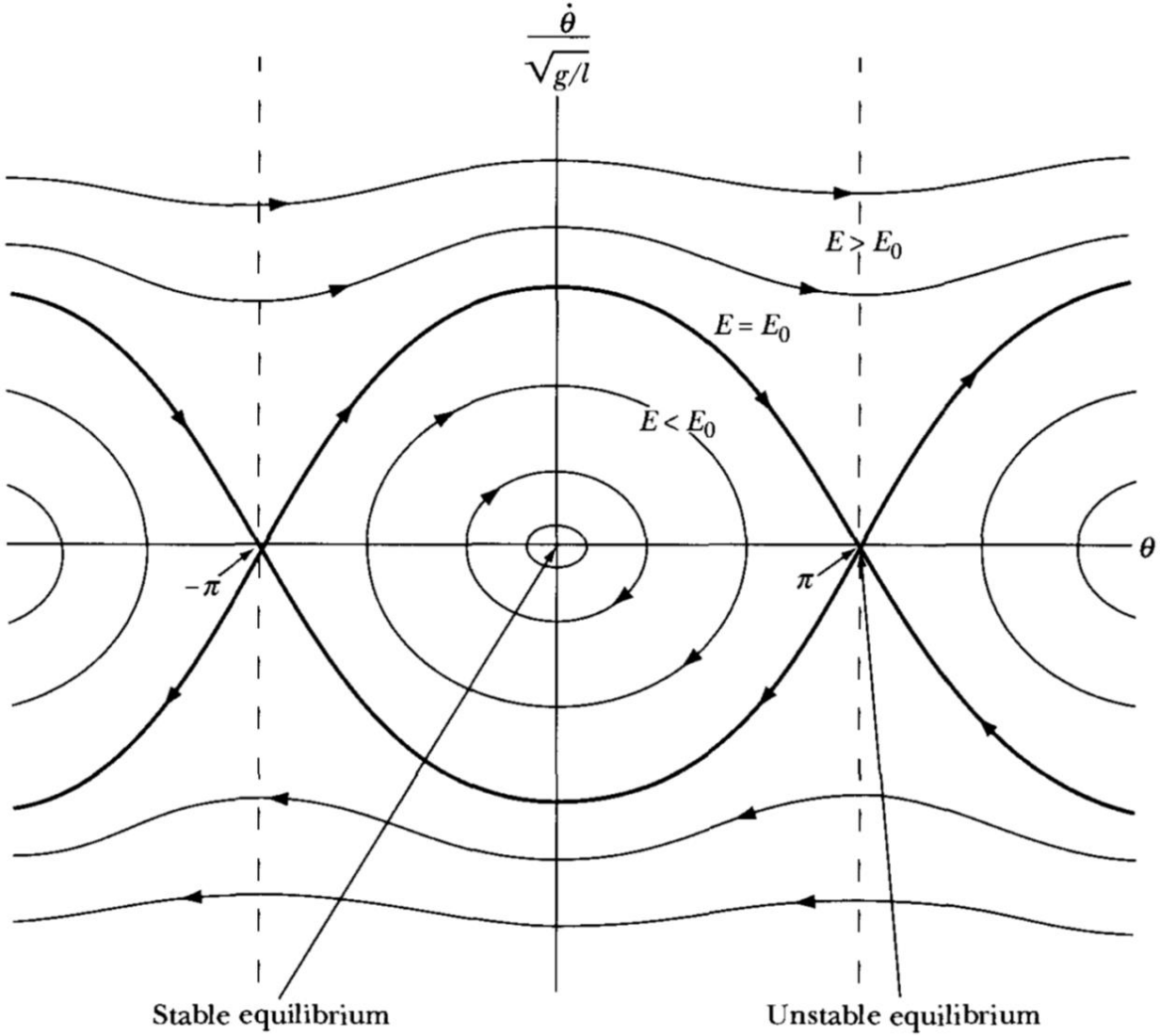


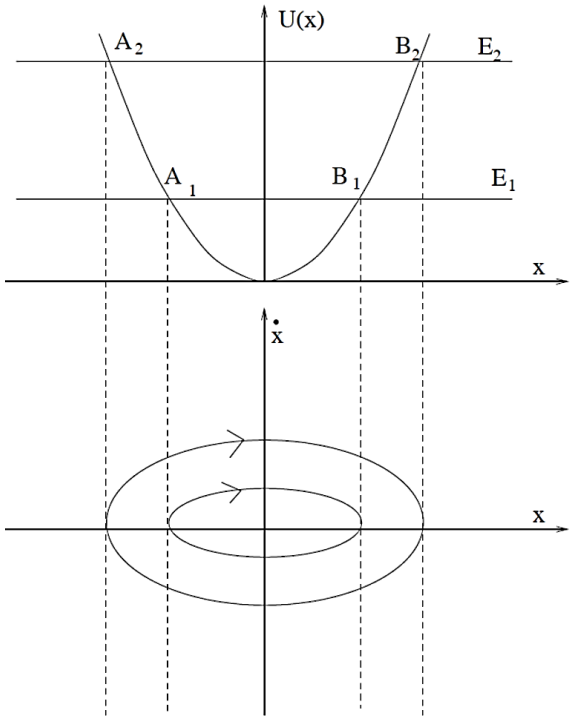
If θ and θ_0 are small

$$\left(\sqrt{\frac{l}{g}}\dot{\theta}\right)^2 + \theta^2 \cong \theta_0^2$$

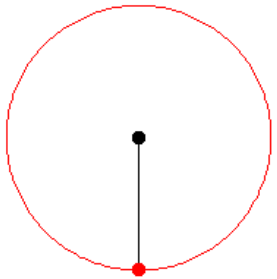
$$E < 2mgl \equiv E_0$$







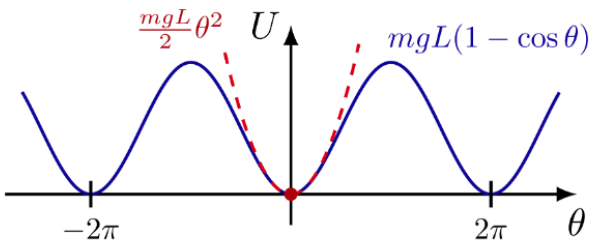
NONLINEAR SYSTEM: PLANE PENDULUM

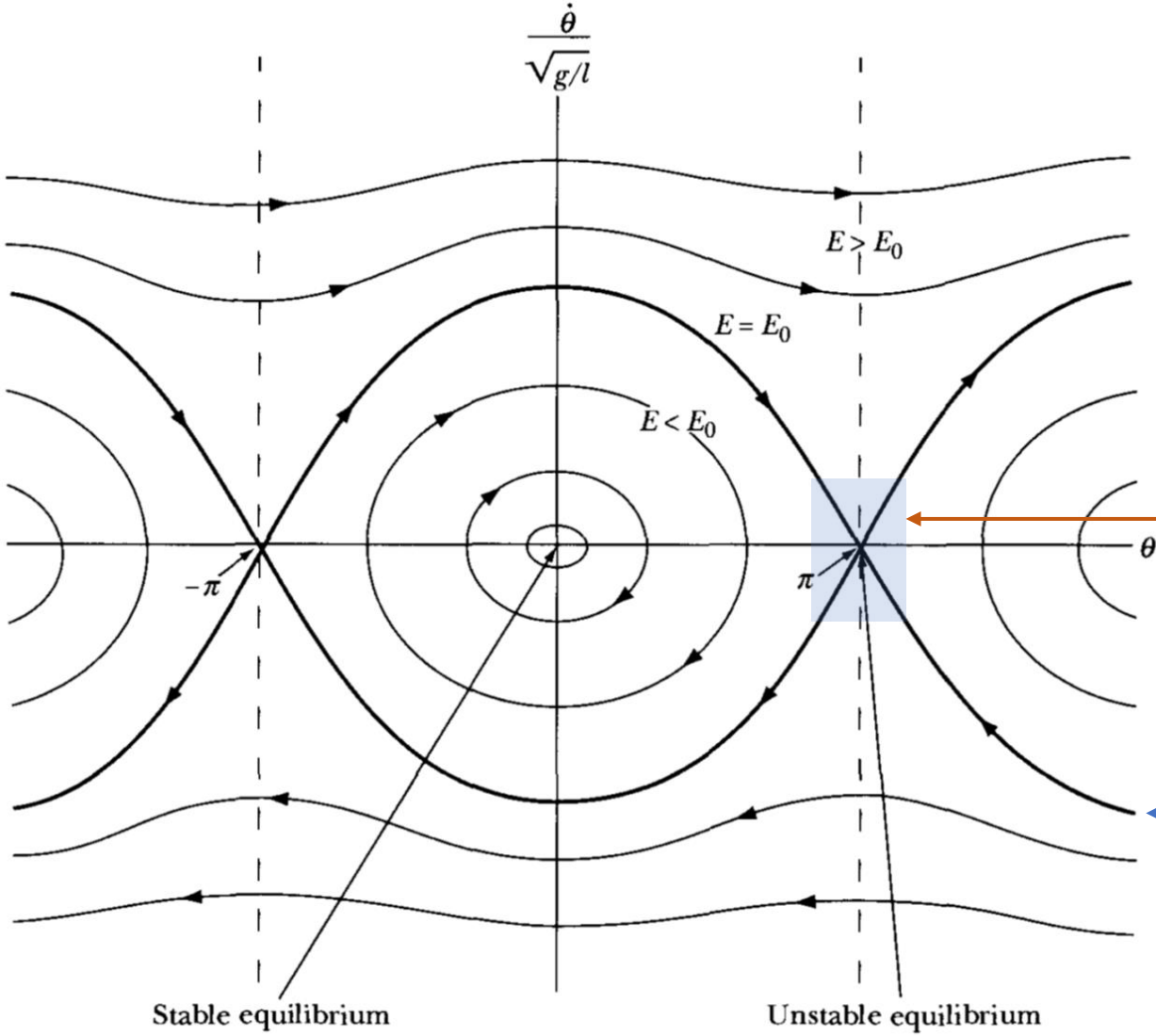


If θ and θ_0 are small

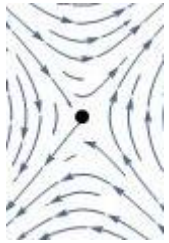
$$\left(\sqrt{\frac{l}{g}}\dot{\theta}\right)^2 + \theta^2 \cong \theta_0^2$$

$$E < 2mgl \equiv E_0$$





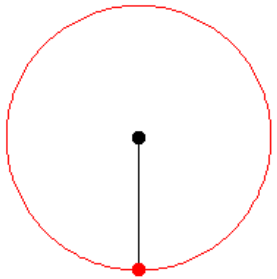
Extremely sensitive to the initial conditions



$$E = E_0$$

$$\dot{\theta} = \pm 2\sqrt{\frac{g}{l}}\cos(\theta/2)$$

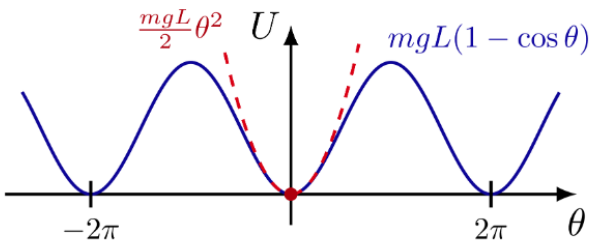
NONLINEAR SYSTEM: PLANE PENDULUM

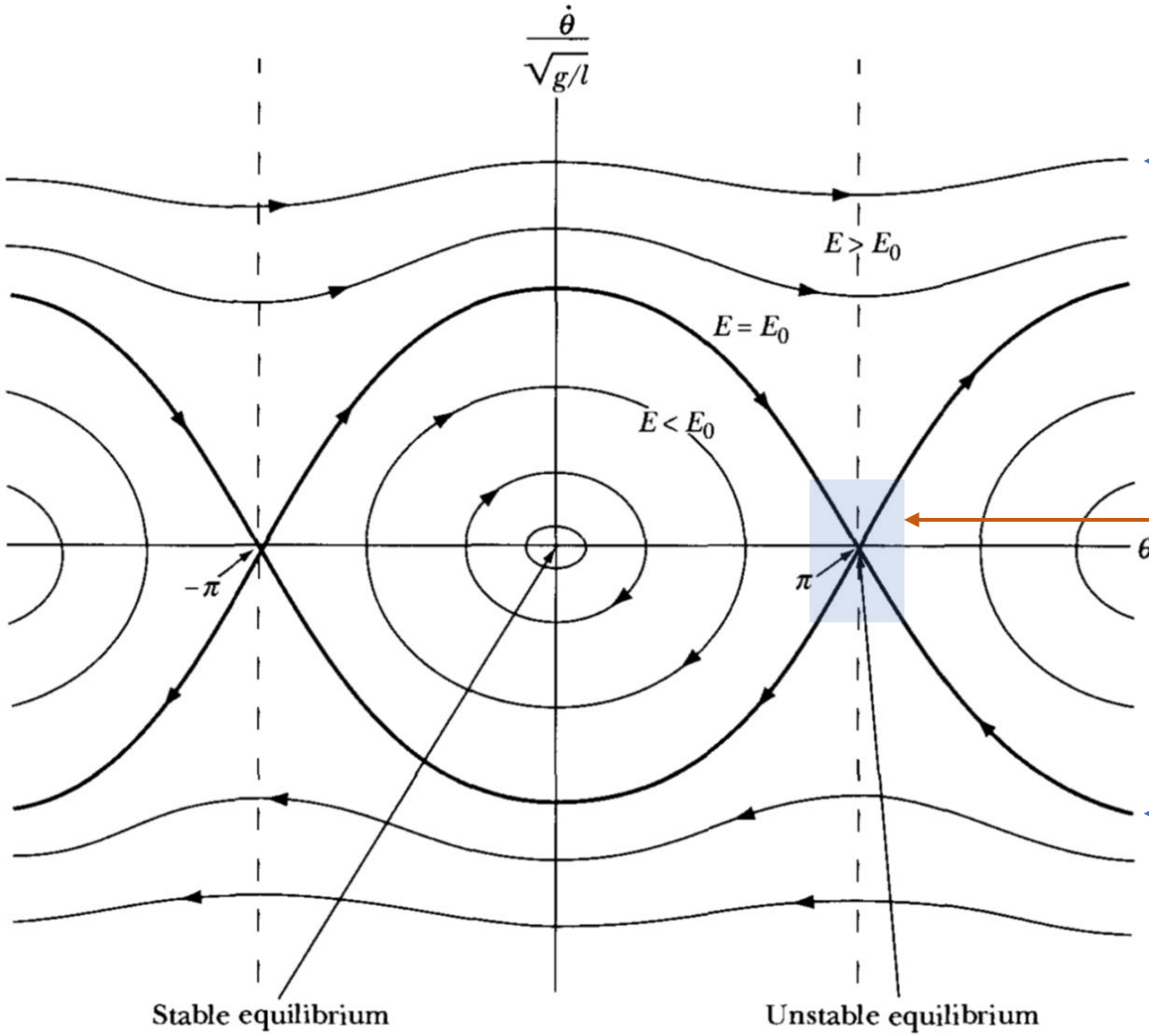


If θ and θ_0 are small

$$\left(\sqrt{\frac{l}{g}}\dot{\theta}\right)^2 + \theta^2 \cong \theta_0^2$$

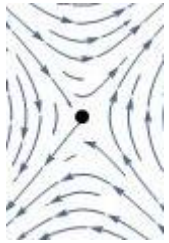
$$E < 2mgl \equiv E_0$$





No oscillations, complete revolution

Extremely sensitive to the initial conditions



$$E = E_0$$

$$\dot{\theta} = \pm 2\sqrt{\frac{g}{l}}\cos(\theta/2)$$

NONLINEAR SYSTEM: PLANE PENDULUM

Damped

